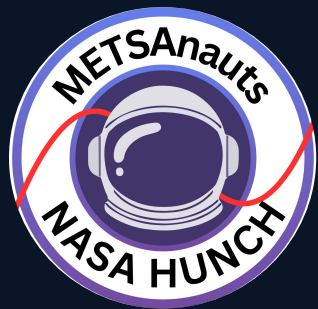
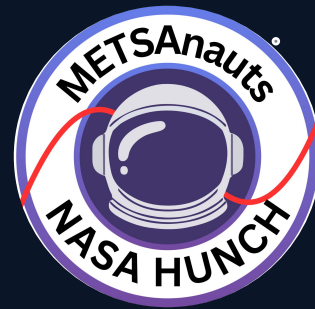




NASA HUNCH

METSAonauts

Ranchview High School



HERA Rover & BothScape Simulation Project

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Our Team

HERA Rover & BothScape Project



Ranchview High School | NASA HUNCH Program | 2026

Our Mission

Problem Statement

The HERA program at NASA is part of the Artemis II series of missions that plan to launch within the next 1-2 years.

Our challenge is to aid the astronauts—teachers, geologists, and specialists—on these missions by designing a set of 4 rovers to traverse the harsh environments of the Moon and Mars, alongside a model landscape that accurately simulates those extraterrestrial conditions.

In practice, these rovers would be used to scout and map the terrain, as well as for the mining of precious metals.

Our Goals

- Build an accurate 8 ft x 8 ft lunar environment where rovers can navigate and carry out tasks.
- Create a physical rover that is able to carry out mining, support, and reconnaissance missions
- Test out the performance of rovers on mock Lunar/Martian conditions
- Immerse astronauts into the experience of a mission.
- Make a user-friendly way to control and adjust the rover

Rover Systems – Overview

Key Design Features

Rocker Bogie Suspension

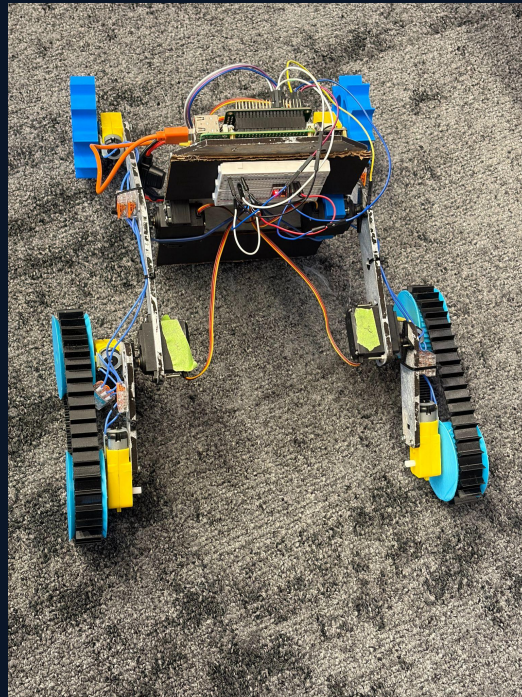
Features a base with active solar tracking. The flexible suspension can handle harsh change in terrain while keeping the rover and its components upright and unharmed.

Half-Tread Wheel System

Increased surface grip for navigating coarse regolith and rocky terrain. Increases stability over traditional wheel systems.

Solar Panel Charging

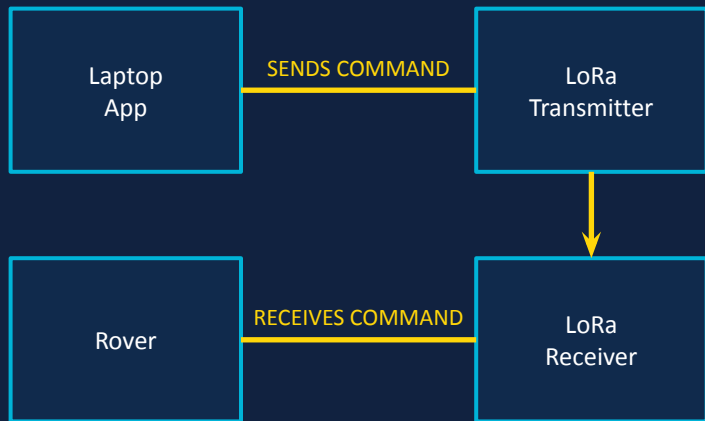
Highly flexible chassis allows simple and efficient recharging via underside solar panels; 60 degree rotation to maximize solar ray capture.



6 independently-motored wheels
Raspberry Pi control system
Robotic claw for sample collection

Communication Systems

Wireless Control Flow



- Commands are packaged for reliable transmission
- Sent wirelessly via long-range radio
- Rover receives and acts on commands automatically

Communication Stack

LoRa

Long-range communication and data transfer when other links are unavailable

LAN

A local wireless network that keeps the rover reliably connected

Bluetooth

Close-up video transfer from the claw-mounted camera

Astronaut-friendly app with intuitive interface for sending predefined and automatically generated commands without technical knowledge.

Mobility & Suspension System

6-Wheel Drive with Half-tread System

- 6 independently-motored wheels give the rover full control across loose, uneven terrain.
- A half-tread design keeps it stable and grounded on more granular surfaces.
- Our wheel prototypes are made out of Creality HyperPLA, a lightweight material that is easy to manufacture and replace.

Suspension System

- A low center of gravity keeps the rover balanced on slopes and uneven surfaces.
- Front wheels climb over obstacles while rear wheels hold the rover steady.
- Weight is distributed evenly across all wheels, no single point takes too much strain.
- High ground clearance increases maneuverability.
- The suspension adjusts for the terrain.

Mobility & Claw System

LDR System

- A Light Dependent Resistor (LDR) on the underside of the rover continuously reads surrounding light levels.
- The Raspberry Pi processes those readings to find where the light is the strongest.
- The rover then repositions itself to face that direction, maximizing solar panel exposure.

Robotic Claw

- Collects loose soil and rocks to simulate mining and sample collection tasks.
- A camera mounted on the claw gives the astronaut a clear view for precise positioning and control of the claw.
- Controlled remotely via either LoRa or the WiFi interface depending on how far the rover is.

BothScape — Simulation Terrain

8 ft × 8 ft landscape | 2 ft × 2 ft terrain sections | Foamboard, foam gap filler, and diatomaceous earth regolith

Moon Region — Shackleton Crater

- Models the lunar south pole with permanent shadow zones
-
- Tests rover navigation in low-light conditions
- Validates power-system behavior under constrained solar input

Mars Region — Ancient Terrain

- Smoother, more eroded terrain reflecting Mars' older surface
- Regolith mimics basaltic volcanic soil with fine, dusty textures
- Includes ancient channel features and lava plains not found on the Moon
- Tests rover endurance on dry, powdery surface conditions

Key Innovations

01

Solar Panel Placement

The solar panels are mounted on the bottom of the rover. Due to the flexible rocker-bogie suspension system, they can tilt upward to catch sunlight even when the sun sits low on the horizon (like on the lunar south pole).

02

LoRa Communication Fallback

Lets astronauts send basic commands to the rover over distances up to ~1km. This acts as a reliable backup when the rover is too far for a WiFi signal to reach.

05

Half-Tread System

The half-tread system Increases grip compared to just 6 independent wheels, helping it stay stable and traverse uneven terrain.

03

Astronaut-friendly UI

An intuitive web app lets astronauts control the rover over Wifi or LoRa with no technical knowledge required. An AI-powered interface is also available to dynamically generate code, allowing for astronauts to use the program easier.

04

LAN Connectivity

A laptop hosts a local area network, allowing with the rover to connect directly to it with no internet needed, keeping communication fast and reliable.

06

Powered Suspension

The rover's flexible suspension lets it shift and adjust its body to match terrain or angle towards the sun, making the rover more adaptable and efficient.

Budget Chart

